ClaimGuard server deployment and test report

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Summary

ClaimGuard runs on your server at https://claimguard.kernelsi.com with every feature on (AI explanations, document reading, search by meaning), and 20 of 26 test measures pass. Run `server_20261031` took about 2 hours on 31 October 2026.

- **No claim was lost.** Five services were restarted under load and every acknowledged claim was still counted (0 lost).
- **Throughput is far above target.** 376 AI explanations a minute (target 200) and 11,721 claims a minute through the pipeline (target 200).
- **Security is sound from outside.** Only ports 80 and 443 are open on the server, old TLS is refused, and anonymous requests get 401.
- **Six measures fail.** Two are test artefacts (port 8080 seen through Cloudflare, a login-failure probe). Four are real work items: photo reading, one rejected batch, one off-topic string accepted, and API memory use. Details in section 4.

Deployment

Item	Value
Server	Ubuntu 26.04 in a Proxmox container, AMD Ryzen 7 5700X (8 threads), 12 GiB RAM, RTX 3080 10 GB
Stack	Docker Compose: API and worker (Rust), MongoDB 7, Redis 7, Caddy proxy, document reader (Tesseract OCR), embedding service (bge-m3, CPU), vLLM serving Gemma 3 4B (fp8) on the GPU
Address	https://claimguard.kernelsi.com through Cloudflare; the server itself listens only on 80 and 443
Version	Updated once to the newer project zip without losing data or settings

Fixes made along the way, so they are not forgotten on the next update:

- **GPU in the container:** GPU libraries copied by version, CDI device access, an extra `/opt/nvrtc13` library mount, and `/tmp` set to `exec` for the model server. These live in `deploy/rust/compose.gpu.yml`.
- **Domain login error:** Caddy now sends the real domain as `Host` to the API (`header_up Host`) and has a default SNI. These live in `deploy/Caddyfile.internal`.
- **Search service:** points at the local bge-m3 model folder because it runs offline.
- **Memory limits:** API and worker raised from 768 MiB to 3 GiB after both were killed for lack of memory.
- **HTTP to HTTPS:** a Cloudflare redirect rule sends port 80 visitors to HTTPS.

Keep `.env`, `Caddyfile.internal` and `compose.gpu.yml` out of any future update copy; the update step must not overwrite them.

Results by experiment

Synthetic claims only. Target and result come straight from the run's files in `outputs/server/server_20261031/`.

Experiment	What it measures	Target	Result	Verdict
S1	AI explanations per minute	at least 200	376.6 (64 at once, 0 errors)	Pass
S2	Claims per minute, whole pipeline	at least 200	11,721	Pass
S2	Batches that failed	0	1	Fail
S2	Slowest read right after load (p95)	under 2 s	1.22 s	Fail (8 requests refused)
S3	Rulebook questions answered with the right rule	at least 90 %	92 % (24 of 26)	Pass
S3	Off-topic strings refused	all	4 of 5	Fail
S4	Document fields read exactly	at least 95 %	86.4 %	Fail
S4	Documents that could not be read	0	0 of 40	Pass
S5	Acknowledged claims lost in restarts	0	0	Pass

Experiment	What it measures	Target	Result	Verdict
S5	Services back after restart	all	5 of 5	Pass
S6	HTTPS, TLS, headers, anonymous access	all yes	7 of 7 yes	Pass
S6	Internal ports closed	yes	8080 reads open	Test artefact
S7	Health not down, audit chain and ledger verify	yes	yes	Pass
S7	No failing health probe	yes	`signals` degraded	Test artefact
S0	Peak memory vs server (12 GiB)	up to 85 %	75 %	Pass
S0	Peak GPU memory vs card (10 GiB)	up to 95 %	77 %	Pass

The run's own scoreboard counts 20 of 26 measures passing. The two "test artefact" rows are why the real position is better than that number; section 4 says why.

Restart test (S5), time the service was unavailable:

Service	Unavailable (s)	Requests refused	Claims acknowledged	Lost
worker	0	0	430	0
api	5.4	20	300	0
redis	0	0	380	0
mongo	0	0	360	0
proxy	0.5	10	195	0

Findings and what to do

Two of the failures come from how the test was run, not from the server. Four are real.

Not server problems

- **S6, port 8080 "open":** the scan went through Cloudflare, which itself answers on 8080. On the server, `ss` shows only ports 80 and 443 listening (Caddy). Nothing to fix.

- **S7, `signals` degraded:** the probe counts failed logins, and the test accounts produced several. It clears on its own after about an hour without failures. Re-run S7 then for a clean pass.

Real findings

- **Photos are read poorly (S4, 86.4 %).** Digital PDFs are 100 %, scans 98.2 %, image-only PDFs 97.3 %, but photos only 50 % (2 of 10 fully right). Only Tesseract was used. A vision model (the `vision` profile) is the planned way to lift photos; until then, ask staff to submit scans, not phone photos.
- **One batch was rejected in the 30-minute steady load (S2).** 1 batch failed out of 2,990 claims, and 8 health-panel requests were refused right after the heavy load. Check the API log around the failure and whether a rate limit caused it.
- **One off-topic string was answered instead of refused (S3).** The cut-off for "too far from any rule" (`EMBEDDER_THRESHOLD=0.485`) can be raised slightly, at the cost of refusing a few real questions. Two real questions also missed their rule ("claim line without a service date", "how many days after the service can a claim be submitted").
- **The API process gets close to its memory limit (S0).** Its peak was 2.77 GiB of the 3 GiB limit (90 %), because the queue held about 8,500 test claims. The default 768 MiB was far too low for that volume. Start from clean data before real use, and keep the 3 GiB limit.

Already fixed during the run: the HTTP to HTTPS redirect (Cloudflare rule), the domain login error, and the memory kills.

Before real use

1. **Cleanup on the server:** remove `ASK_PER_MINUTE` from `.env` (restores the normal question rate limit), recreate the API container, and delete `~/bench.env`, `~/claimguard-new.zip` and `~/update-tmp`.
2. **Start clean:** decide between a fresh database (drops the roughly 8,500 test claims and every account; then create your real admin with `cg admin create-admin`) or keeping the current data and deleting the benchmark accounts CG-4002, CG-4003 and CG-1001 from the console.
3. **Keep the backups private:** `backup-before-update/` (it holds the encryption key) and the `.env` copy are now on your PC; store them off the Downloads folder, and never in a zip you share.
4. **Security housekeeping:** delete the Hugging Face token you no longer need, and keep SSH (port 22) closed to the internet.
5. **Re-test later:** run S7 after an hour without failed logins, and S4 again once a vision model is added.

Untested so far: real hospital paper, handwriting and Arabic documents. Everything here used synthetic claims.